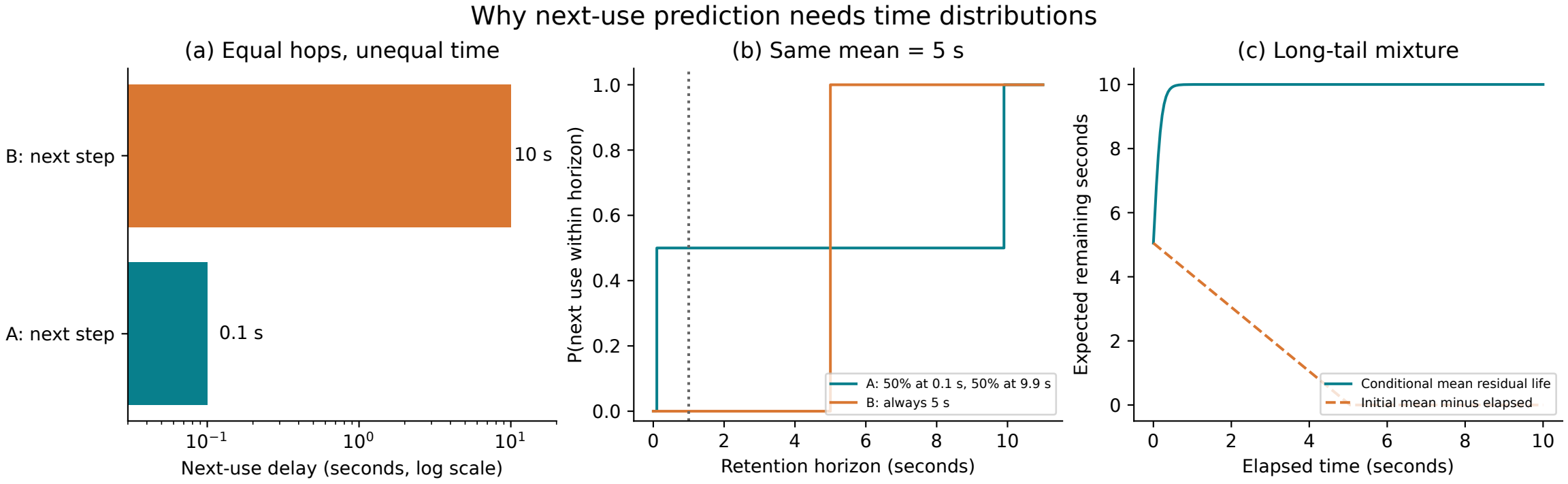
Why next-use prediction needs time distributions
(a) Equal hops, unequal time
B: next step
10 s
A: next step
0.1 s
10^{-1}
10^{0}
10^{1}
Next-use delay (seconds, log scale)
(b) Same mean = 5 s
P(next use within horizon)
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
Retention horizon (seconds)
A: 50% at 0.1 s, 50% at 9.9 s
B: always 5 s
(c) Long-tail mixture
Expected remaining seconds
10
8
6
4
2
0
Elapsed time (seconds)
Conditional mean residual life
Initial mean minus elapsed